

# Luca Santarelli

## Software Engineer

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Portfolio: <https://luca-santarelli.onrender.com/> | GitHub: [github.com/ghostbustersrock](https://github.com/ghostbustersrock) | LinkedIn: [linkedin.com/in/luca-s-a68182118](https://linkedin.com/in/luca-s-a68182118)

Work authorisation: UK Indefinite Leave to Remain (settled status); Italian/EU citizen

## PROFESSIONAL SUMMARY

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Software Engineer experienced in building and maintaining production backend systems, data platforms and AI-powered internal tools. Passionate about designing reliable software that automates business processes, improves operational efficiency and delivers measurable value. Experienced across Python, cloud infrastructure, data engineering and modern web technologies, with a strong focus on ownership, scalability and continuous improvement.

## TECHNICAL SKILLS

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**Languages & Backend:** Python, FastAPI, Flask, REST APIs, SQLAlchemy

**Frontend:** React.js, Vue.js, JavaScript, HTML, CSS

**Databases & Data:** PostgreSQL, ETL Pipelines, Data Engineering, Pandas, NumPy

**Cloud & DevOps:** AWS (EC2, ECS, S3), Docker, GitLab CI/CD

**AI:** Large Language Models (LLMs), Claude / ChatGPT Integration, MCP Servers, Prompt Engineering

**Testing & Tools:** Pytest, Postman, Git, Jira, Confluence

## EXPERIENCE

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### Software Engineer

June 2024 - Present

#### AlliedOffsets, London

- One of two software engineers responsible for designing, maintaining and evolving the company's core backend platforms supporting carbon market operations.
- Designed, maintained and continuously improved production ETL pipelines responsible for ingesting and validating partner datasets.
- Delivered production-ready backend services and full-stack internal applications using Python, FastAPI, React and PostgreSQL, supporting critical business operations.
- Designed and implemented AI-powered internal tooling, including an MCP server, using Large Language Models to improve knowledge discovery and automate internal workflows.
- Built multiple greenfield products from initial concept through production deployment, collaborating closely with stakeholders to translate business requirements into software solutions.
- Investigated and resolved production incidents, ensuring reliability and stability across business-critical systems.
- Automated manual operational processes through custom internal tooling, reducing repetitive work and improving team efficiency.
- Worked directly with external partners and internal stakeholders to deliver technical solutions to operational challenges.

### Software Engineer

Nov 2021 - Sep 2022 & Nov 2023 - Mar 2024

#### Hullabalook, London

- Developed customer-facing web applications using Vue.js with JavaScript and TypeScript, using Python to process customer datasets supporting application functionality.
- Modernised legacy applications, improving long-term maintainability and reducing technical debt.
- Collaborated across engineering and product teams to deliver reliable software enhancements.

### Python Developer

Sep 2022 - Nov 2023

#### QuantSpark, London

- Designed and implemented backend APIs supporting financial analytics platforms.
- Developed scalable backend services using Flask, SQLAlchemy and PostgreSQL.
- Processed and analysed large datasets using Python, Pandas and NumPy.
- Containerised applications using Docker and contributed to automated testing with Pytest.

- Collaborated with consultants and clients to deliver reliable software solutions.

## FEATURED PROJECTS

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### Maison Scanner

<https://maison-scanner.com/>

Designed and built a property search platform that aggregates rental listings from multiple UK property websites using automated web scraping and API integrations. Fully hosted on a self-managed personal server.

- Automated data collection using Selenium and BeautifulSoup.
- Integrated REST APIs to enrich search results and implemented Redis caching to improve performance.
- Implemented and continue to refine a solution for detecting and reducing duplicate listings to improve search quality.

### Personal Engineering Portfolio

<https://luca-santarelli.onrender.com/>

Designed and developed a modern engineering portfolio showcasing professional projects, technical skills and engineering philosophy. Built as a long-term personal platform demonstrating software engineering experience and continuous learning.

## EDUCATION

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### MSc Computer Science - Distinction

Sep 2020 - Sep 2021

[Queen Mary University of London](#)

### BEng (Hons) Electrical and Electronic Engineering - First Class Honours

Sep 2017 - Jul 2020

[Queen Mary University of London](#)

## CERTIFICATIONS

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Learn React - Scrimba

December 2025

Investing in Stocks: The Complete Course - Udemy

January 2025

The Complete Python Developer - Zero To Mastery

December 2023

Responsive Web Design - freeCodeCamp

December 2021

## LANGUAGES

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Italian (Native), English (Fluent), Spanish (Intermediate), French (Beginner/Intermediate), Portuguese (Beginner)